

JEE MAIN REVISION MASTER: THERMODYNAMICS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 14 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	First Law, Enthalpy & Hess's Law	10 PYQs	Thermochemistry
Moderate	Entropy, Gibbs Free Energy & Spontaneity	4 PYQs	Thermodynamic Spontaneity

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
First Law, Enthalpy & Hess's Law	$\Delta H = \Delta U + \Delta n_g RT, \quad \Delta n_g = \sum n_{g,\text{prod}} - \sum n_{g,\text{react}}, \quad \Delta H_r = \sum \Delta H_f^\circ(\text{react}) - \sum \Delta H_f^\circ(\text{prod})$
Entropy, Gibbs Free Energy & Spontaneity	$\Delta G = \Delta H - T\Delta S, \quad \Delta G^\circ = -2.303 RT \log_{10} K, \quad \Delta S_{\text{universe}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 First Law, Enthalpy & Hess's Law (High Yield — 10 PYQs)

Core Principle: First Law $\Delta U = q + w$. Expansion work against constant pressure $w = -P_{\text{ext}}\Delta V$. Relation between ΔH and ΔU : $\Delta H = \Delta U + \Delta n_g RT$. Hess's Law of constant heat summation: overall enthalpy change is independent of intermediate steps. Standard enthalpy of formation of elements in their reference state is defined as zero.

■ **Caution / JEE Pitfall:** When calculating Δn_g , count ONLY gaseous species; completely ignore solids and pure liquids!

■ **Speed Trick:** For reactions with $\Delta n_g = 0$ (e.g. $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$), ΔH is exactly equal to ΔU .

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A gas can be taken from A to B via two different paths ACB and ADB. When path ACB is used, 60 J of heat flows into the system and 30 J o..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Which of the following is a correct statement for a thermodynamic system?..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "Which one of the following reactions has $\Delta H = \Delta U$?..." [Ans: N/A]

3.2 Entropy, Gibbs Free Energy & Spontaneity (Moderate Yield — 4 PYQs)

Core Principle: Gibbs-Helmholtz equation: $\Delta G = \Delta H - T\Delta S$. Criteria for spontaneity at constant T and P: $\Delta G < 0$ (spontaneous), $\Delta G = 0$ (at equilibrium), $\Delta G > 0$ (non-spontaneous). Spontaneity temperature threshold: $T = (\Delta H / \Delta S)$. Relation to equilibrium: $\Delta G^\circ = -RT \ln K = -2.303 RT \log K$.

■ **Caution / JEE Pitfall:** A reaction with $\Delta H > 0$ (endothermic) can still be spontaneous if $\Delta S > 0$ at sufficiently HIGH temperatures ($T > \Delta H / \Delta S$).

■ **Speed Trick:** If $\Delta H < 0$ and $\Delta S > 0$, ΔG is negative at ALL temperatures \implies always spontaneous.

Representative Source PYQs:

- **[2026 Shift] Q#3 (MCQ):** "A: Entropy of a perfect crystalline solid at absolute zero approaches zero. B: For spontaneity of a reaction, $T\Delta S > \Delta H$. Among t..." **[Ans: N/A]**
- **[2023 Shift] Q#2 (MCQ):** "Temperature of 25°C in Fahrenheit and Kelvin scale respectively are..." **[Ans: N/A]**
- **[2020 Shift] Q#4 (MCQ):** "When the same quantity of heat is absorbed by a system at two different temperatures $T_{(1)}$ and $T_{(2)}$, such that $T_{(1)} > T_{(2)}$, change ..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A gas can be taken from A to B via two different paths ACB and ADB. When path ACB is used, 60 J of heat flows into the system and 30 J of work is done by the system. If path ADB is used, work done by the system is 10 J. The heat flow into the system in path ADB is

- (A) 80 J
- (B) 20 J
- (C) 100 J
- (D) 40 J

Correct Answer: None

Solution Analysis: By the first law of thermodynamics, $\Delta U = Q - W$. Here, Q is the heat supplied to the system, and W is the work done by the system. For path ACB: Heat added, $Q_{(ACB)} = 60$ J Work done, $W_{(ACB)} = 30$ J So, $\Delta U = 60 - 30 = 30$ J The change in internal energy depends only on the initial state A and final state B. It does not depend on the path followed. So, for path ADB also, $\Delta U = 30$ J For path ADB: Work done, $W_{(ADB)} = 10$ J Using the first law again, $\Delta U = Q_{(ADB)} - W_{(ADB)}$ $30 = Q_{(ADB)} - 10$ $Q_{(ADB)} = 30 + 10 = 40$ J The heat flow into the system in path ADB is 40 J.

Benchmark Problem 2: KCET 2026 (Q#3)

A: Entropy of a perfect crystalline solid at absolute zero approaches zero. B: For spontaneity of a reaction, $T\Delta S > \Delta H$.

Among the two statements given above, identify the correct answer from the options given below.

- (A) Both 'A' and 'B' are true
- (B) 'A' is true but 'B' is false
- (C) Both 'A' and 'B' are false
- (D) 'A' is false but 'B' is true

Correct Answer: None

Solution Analysis: Statement A is true. According to the third law of thermodynamics, the entropy of a perfectly crystalline solid at absolute zero, that is at 0K, is taken as zero. So, $S=0$ at 0K for a perfect crystalline solid. Statement B is also true.

For a reaction to be spontaneous at constant temperature and pressure, $\Delta G < 0$. We know, $\Delta G = \Delta H - T\Delta S$. For spontaneity, $\Delta H - T\Delta S < 0$. So, $T\Delta S > \Delta H$. Hence, both statements A and B are true. The correct answer is: Option A